

Read This First

Your private, outsourced Damodaran valuation system, explained in plain words

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1 How to use this document

Read it once, front to back, before you open Cursor or create any account. It is written for someone who is new to software tooling and is relearning valuation. Every term is defined the first time it appears, and there is a glossary at the end. When a section says “you will see”, that is literally what will appear on your screen.

The companion documents are:

- **Damodaran, the important parts** (damodaran-essentials.pdf): your revision brain-dump. Read it second, before anything technical.
- **Teaching Guide** (guide.pdf): where and how to learn each tool, one chapter per tool, in the order the project needs them.
- **The Other Side** (other-side.pdf): what Damodaran got wrong, who bet against him and won, and what the academics say. Read it after the essentials so the method never becomes a religion.

The diagrams in this document are also live boards in your Figma account (FigJam). Open them, drag boxes around, add notes. They are yours.

Board	Link
System architecture	https://www.figma.com/board/DtFTvDZeSQpuKLM5tavDUT
Nightly data flow	https://www.figma.com/board/6LSfZZfWAnbQVY0twr7OMw
The valuation chain	https://www.figma.com/board/SNaMV0lrKHeDRQgelrbPsp
The memo pipeline (story to numbers)	https://www.figma.com/board/LGcrXp6yT3fh4O2IV3mEY8
The milestone map	https://www.figma.com/board/LXcGUT7C2ArsRkwsUvRBWt

2 1. What we are building, in one page

You used to value one company at a time in Damodaran’s spreadsheet. You want a machine that does that for many companies, in India and the United States, with his lectures on tap, private to you, and with nothing running on your Mac except an editor. It has seven parts.

1. **A calculator.** A program that takes a company’s numbers (revenue, operating income, debt, cash, shares) plus Damodaran’s reference tables (industry betas, country risk premiums, the risk-free rate) and computes value per share exactly the way his spreadsheet does. His spreadsheet ships with a worked example: Almarai, a Saudi food company, value 7.187840270062114 per share against a price of 72.28. Our program must reproduce that number to six decimal places before it is allowed to value anything else. We call that the **golden test**.
2. **A database in the cloud (Neon).** Every number the calculator uses is stored with three labels: where it came from (source), whether you may show it to other people (licence class), and which

snapshot of Damodaran's data it belongs to (vintage). The database refuses to save a number that is missing any of the three.

3. **A librarian in the cloud (Pinecone and Voyage).** His 609 blog posts, lecture packets, book, datasets and transcripts are cut into about 60,000 passages. Each passage is turned into a list of numbers (a vector) so that a question can find the passages that talk about the same thing. Answers cite the passages they used.
4. **A memo writer.** Before the calculator runs on a company, a model reads the company's latest filing in one go, fetches the relevant Damodaran passages from the librarian, and writes a "story to numbers" memo in his own template. Every claim in the memo is labelled possible, plausible or probable. The memo says what the current price already assumes (a reverse DCF), names the kind of uncertainty the company carries (young, distressed, cyclical, financial, complex, emerging market) and proposes a value for every calculator input, with a citation to a filing page or a Damodaran passage. You review it, change what you disagree with, and approve. Only then does the calculator run. A second, opposing memo is required for every company.
5. **Robots that run at night (GitHub Actions).** Scheduled jobs pull new filings from the US regulator (SEC EDGAR, free), later from a paid Indian data feed (EODHD), re-value your watchlist, check whether Damodaran posted a new risk-premium file, pull licensed news headlines, and write everything to the database. Once a night is the right pace. Nothing here is intraday.
6. **A private dashboard (Hex).** A hosted notebook over the same database shows watchlist diversification by sector and country, value-versus-price distributions, Monte Carlo ranges, an accuracy scoreboard (how past valuations did against later prices) and a descriptive "is the market on the rails" panel built from Damodaran's own implied risk-premium series. As your degree adds ideas, you add notebooks against the same tables.
7. **Your daily tool: Cursor.** You already pay for it. The system's tools plug into Cursor so you can type "value Trent", "write the memo for Apple" or "ask Damodaran why he strips the default spread out of the Indian risk-free rate". Cline, a free extension, sits in the same window as a bench for trying open-weight models. PDFs are only for reading offline.

3 2. Where the pieces live and who runs them

Everything except your editor runs on a managed service. You are the client; they are the contractors. Here is each contractor, what it does in one line, and why it was chosen.

Contractor	Job	In one line	Why this one
GitHub	Holds the code; runs the robots	A private folder in the cloud with a history of every change, plus computers you can rent by the minute to run scheduled jobs	Where the code has to live anyway; 2,000 free minutes a month on a private repo
Neon	The database	Managed Postgres that switches itself off when idle, with a copy of your whole database per change request	Idle costs nothing; branches make experiments safe

Contractor	Job	In one line	Why this one
Pinecone	The librarian's index	Stores the 60,000 passage vectors and finds the nearest ones to a question	Free at this size; purpose-built
Voyage	Turns text into vectors, reranks results	The model that understands "sales to capital" and "reinvestment efficiency" are neighbours	200 million free tokens; embeds each passage aware of its neighbours, which rescues the unpunctuated transcripts
OpenRouter	One door to hundreds of AI models	One key, one bill; you swap a model by changing a name	Only router that can guarantee your prompts are not retained
Cloudflare R2	Raw files	The bucket that holds the spreadsheets, zips and OCR output, each named by its fingerprint (hash)	10 GB free, no charge to read files back
Langfuse	Logs of every model call	Which model, which prompt, what it cost, what it returned	Free tier includes evaluations; open source
Hex	The dashboard	SQL and Python notebooks over Neon that turn into private dashboards	The only tool in budget with real Python and product-grade charts
Mistral	OCR, one-off	Turns three image-only lecture decks into text	Best independent score, about a dollar for the job
EODHD	Indian statements (from milestone 4)	Income statements, balance sheets and cash flows for NSE and BSE companies	Only vendor with documented long-history India data
Cursor	Your editor and agent	A VS Code with an AI that plans, edits and shows you every change as a diff	Already paid; safest for a beginner

4 3. The six rules that make the numbers trustworthy

These are the rules that survive every redesign. They are about correctness, not style.

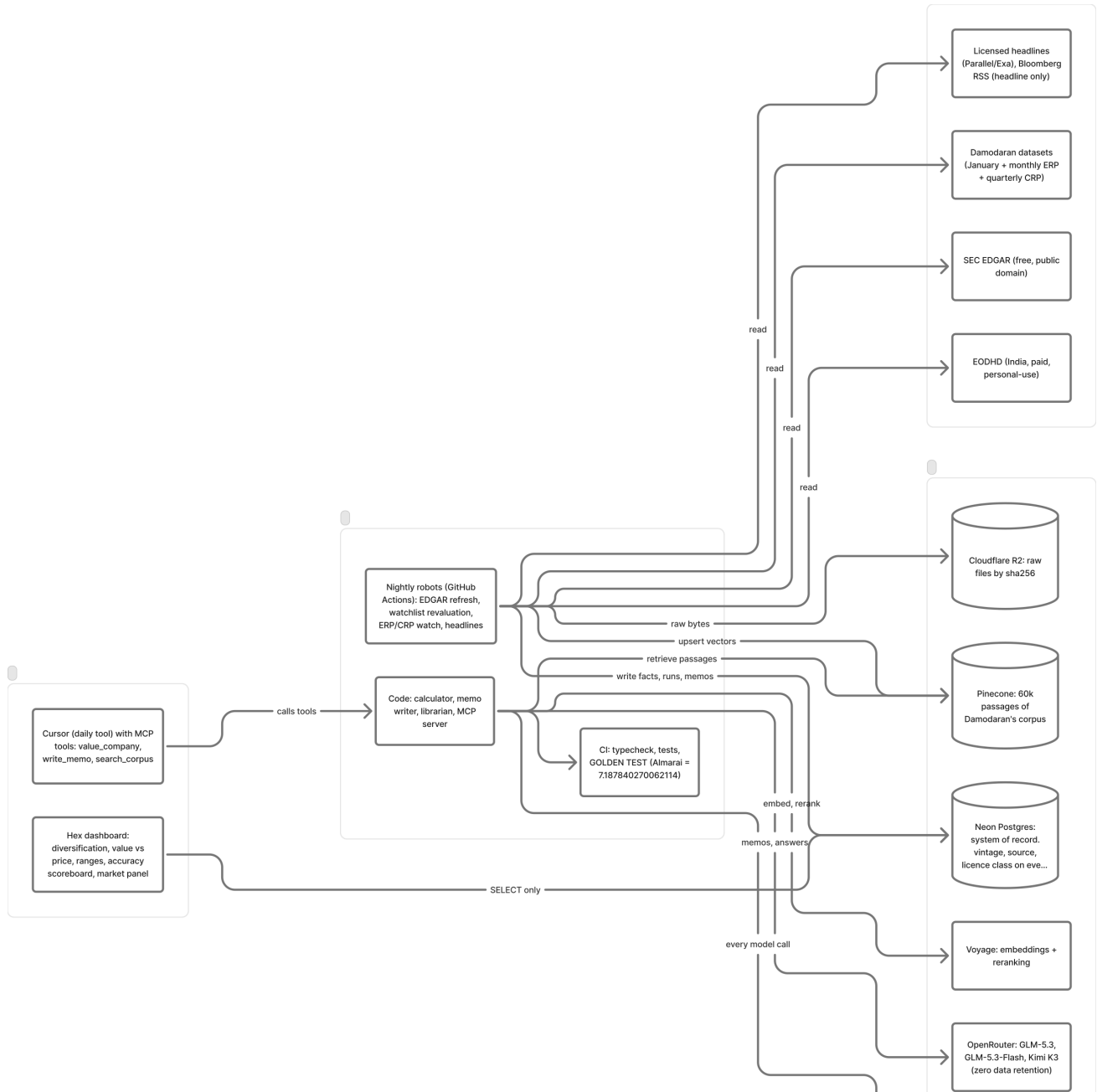


Figure 1: System architecture

4.1 3.1 The golden test

Damodaran's spreadsheet contains a finished example (Almarai). Our calculator must reproduce its value per share, 7.187840270062114, within 0.000001. The expected numbers are pulled out of the workbook by a script, never typed by hand, and the file that holds them is protected so the AI tool cannot "fix" the test instead of the code. Every code change runs this test automatically. If it fails, the change is refused.

What it does and does not prove: it proves the *arithmetic* is faithful, including how an industry beta is re-levered for the company's debt. It says nothing about whether the *inputs* are right. That is why every valuation also carries a range (part 3.5).

4.2 3.2 Vintage

Damodaran refreshes his tables every January. Two of them refresh more often: the country risk premiums quarterly, the implied equity risk premium monthly. So a single year contains several versions of the risk-free rate and the risk premium. In the 2026 data alone there are three risk-free rates (3.95%, 4.58%, 4.75%) and three mature-market risk premiums (4.20%, 4.23%, 4.46%). Mixing a January industry table with a September risk premium produces a confidently wrong answer with no warning. So every valuation run stores three ids: which market file, which country-risk file, which industry file. The database refuses a run without all three.

4.3 3.3 Source and licence class

Every stored fact records where it came from and one of seven licence classes:

Class	Examples	May it appear in the librarian's index?	May it ever be shown to others?
public_domain	SEC EDGAR filings	yes	yes
damodaran_public	his datasets and posts	yes	yes, with attribution
open_access	RBI yields, arXiv, SSRN author copies	yes	yes
asr_youtube	auto-captions of his lectures	yes, marked as approximate wording	quote carefully
licensed_personal	EODHD, your broker's API	no	no, until a commercial licence
proprietary_personal	your Bloomberg reading notes	no	never
link_only	books, paid newsletters	no (we store the link)	link only

A database rule refuses the last three from the index. That rule is what lets you go public later without an audit: "can I show this?" becomes a query.

4.4 3.4 The India rules

- The Indian risk-free rate is the 10-year government bond yield **minus** India's default spread. Otherwise sovereign risk is counted twice, because the cost of debt already adds it.

- Equity risk premium for an Indian company = mature-market premium + India's country premium. January 2026: 4.23% + 2.85% = 7.08% (rating Baa3, default spread 1.87%). July 2026: 6.92%.
- Damodaran publishes industry averages for India, but 27 of the 94 industries have fewer than 10 Indian companies in them. When that happens the system uses the emerging-markets table, then the global table, and prints which one it used.

4.5 3.5 Ranges, not points

Every valuation also produces a sensitivity table and a Monte Carlo range drawn from Damodaran's own industry quartiles. The scoreboard (part 6) later measures the bias and the spread of those estimates against realised prices. Exact engine, uncertain inputs, measured error.

4.6 3.6 No advice

A test scans every report, memo and chat answer for buy, sell, hold or target price. The system prints estimates (value, price as a percentage of value, the probability that value is below price, what the price already requires, and how past calls did). It never tells you what to do.

5 4. The two places an AI model is used

This confused you, so here it is slowly.

Place 1: while you write code. That is Cursor. Cursor includes its own pool of models in the subscription you already pay. You do not touch that. You do not paste any API key into Cursor: doing so is unofficial, switches off half its features, and voids Cursor's promise that your code is not retained.

Place 2: inside the product, when it runs. Our own code calls a model to write memos, answer questions from the librarian, and clean transcripts. Those calls go through OpenRouter with zero-data-retention routing on and prompt logging off. Here the open-weight models go, because this is where the volume and the cost are, and because a model pinned to an exact checkpoint keeps a 2026 valuation reproducible in 2029.

"Open-weight" means the model's weights are downloadable by anyone and hosted by many companies, so no single vendor can switch you off. They are 7 to 40 times cheaper per token than the frontier models. The cost is capability: on the Artificial Analysis Intelligence Index v4.3 (7 September 2026) the frontier scores 53 and the best open-weight model (GLM-5.3) 45. The gap sits in long autonomous coding tasks, not in the "read a filing and extract" work that dominates a valuation pipeline.

5.1 4.1 The model table, explained line by line

Job	Model (OpenRouter id)	Price per million tokens, in / out	Why
Write memos, answer questions	z-ai/glm-5.3	about \$0.92–1.40 / \$3.14–4.40	Best open-weight model on the index (45) and the lowest hallucination rate among strong open weights

Job	Model (OpenRouter id)	Price per million tokens, in / out	Why
Bulk cleaning (punctuating transcripts, tagging passages)	z-ai/glm-5.3-flash:batch	\$0.075 / \$0.25	Half price for non-urgent work; the whole transcript job costs about \$11 once. Reasoning must be switched off or “thinking” tokens are billed as output
Judge (grading the librarian’s answers in evaluations)	moonshotai/kimi-k3	\$2.65 / \$13.28	Best-calibrated open-weight model (it abstains rather than invents); the judge must never be the model being judged
Fallback	deepseek/deepseek-v4.1-flash	\$0.15 / \$0.60	Fastest in class, permissive licence
Never	DeepSeek V4 Pro		94% hallucination rate on the knowledge test, disqualifying for a citation-bearing chat

How to read the columns: “tokens” are word pieces (roughly 4 characters). “In” is what you send, “out” is what the model writes. A typical librarian answer sends about 8,000 tokens and gets 700 back, so it costs about \$0.0143 on GLM-5.3. The model ids and prices are re-checked monthly and pinned in one file; a swap is one line.

6 5. What “scales”, in plain words

You asked whether this design scales. Here is what scales for free and what does not.

Scales for free: the calculator (a pure function; a thousand companies is a thousand calls), the librarian (a question costs the same whether the index holds 30 million or 300 million tokens, because it only fetches 8 passages), the robots (GitHub can run up to 256 jobs at once), the dashboard (it only reads).

Does not scale by itself, and the fix: the database’s single write endpoint (write in batches, never one row at a time); the regulator’s rate limit (about 10 requests per second, so the robots only fetch a company when a *new* filing exists, never everything every night); the Indian data vendor’s call budget (weekly refresh, keyed on their “last updated” field).

Going from 50 companies to 5,000 changes the shape of the nightly job and the database tier. It does not change the calculator, the librarian, the memo writer or the dashboard.

7 6. Estimates, not advice: what the scoreboard shows

Every valuation is stored with a timestamp, its inputs, its memo version and its three vintages. After 90, 180 and 365 days a job records the realised price. The scoreboard then shows, by sector, region and vintage: the average signed error of value versus later price (bias) and its spread (variance), plus the share of calls where value and price moved the way the valuation implied. The banner on that page reads: “The engine is tested (Almarai 1e-6). The forecast is not.”

On “is it working for the next month”: Damodaran’s own data says no valuation method forecasts next year’s return well, and one month is noise. The honest monthly question is “is the method working so far”, and that is what the scoreboard answers as valuations age.

8 7. The nightly robots

At about 6:17 UTC every night, a GitHub Actions job wakes up. For each company on the watchlist it checks whether the regulator lists a new filing. If yes, it downloads the structured numbers, builds the trailing-twelve-month figures (last annual report minus the prior-year interim plus the current interim), writes them to Neon with their source, licence class and vintage, assembles the calculator inputs, runs the calculator, and stores the run. It writes a heartbeat row; if the heartbeat goes stale for 36 hours, you get an alert. Monthly and quarterly, another job checks whether Damodaran posted a new risk-premium or country-risk file; if the file’s fingerprint changed, it loads it as a *new* vintage and opens a GitHub issue for you to decide whether to adopt it. A third job pulls licensed headlines and stores them as annotations only.

9 8. The valuation chain

Left to right: revenue growth drives revenues; the operating margin converges to its target; taxes fade from the effective to the marginal rate; reinvestment is the change in revenue divided by the sales-to-capital ratio, one year behind; free cash flow to the firm is operating income after tax minus reinvestment; the cost of capital comes from the risk-free rate, a re-levered industry beta, the risk premium (mature plus country) and a synthetic credit rating; the terminal value assumes growth at the risk-free rate and returns fading to the cost of capital; failure probability truncates the going-concern value; then debt, minorities, cash, non-operating assets and options are added or removed to reach equity and value per share. The full walk-through with Almarai’s numbers is chapter 3 of “Damodaran, the important parts”.

10 9. The memo pipeline

This is the part that answers your “for every stock there should be rational long-context reasoning”. The model reads the filing whole, pulls the Damodaran passages that apply, writes the story in his template, labels every claim (possible, plausible, probable), works out what the price already requires, names the uncertainty type and proposes each input with a citation. Guard rails reject an uncited claim or an input outside Damodaran’s published plausibility bands (for example, 80% of US companies have a cost of capital between 5.26% and 9.88%). You approve or edit the diff in Cursor. Numbers are written as rows, never parsed from prose, so the calculator is never touched by the model.

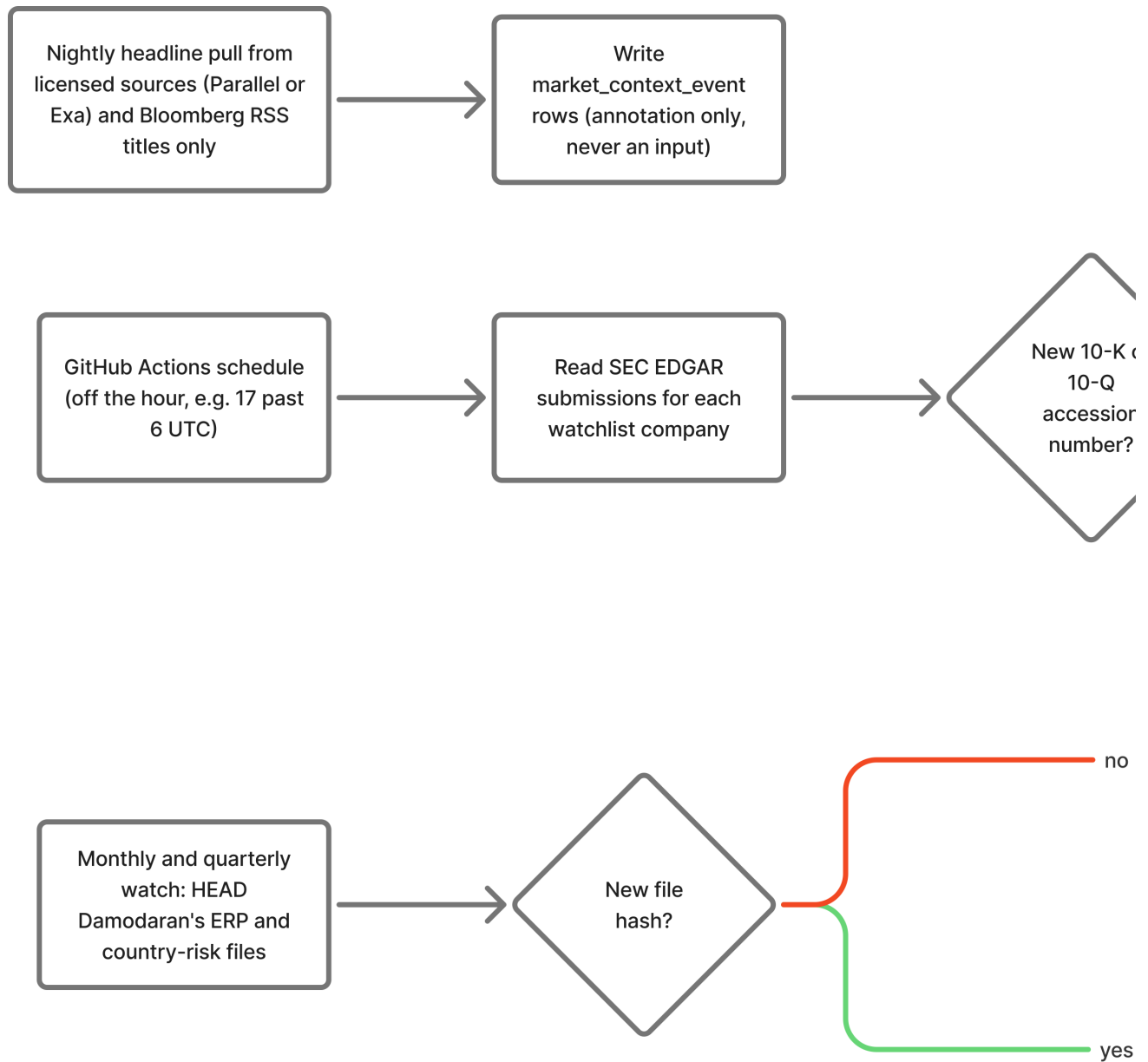


Figure 2: Nightly data flow

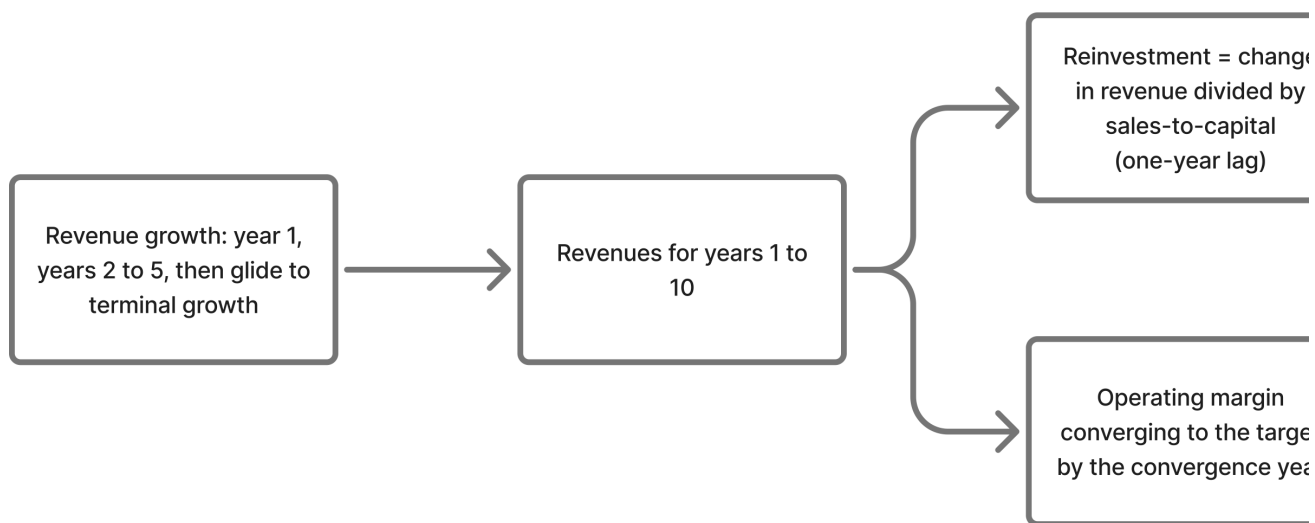


Figure 3: The valuation chain

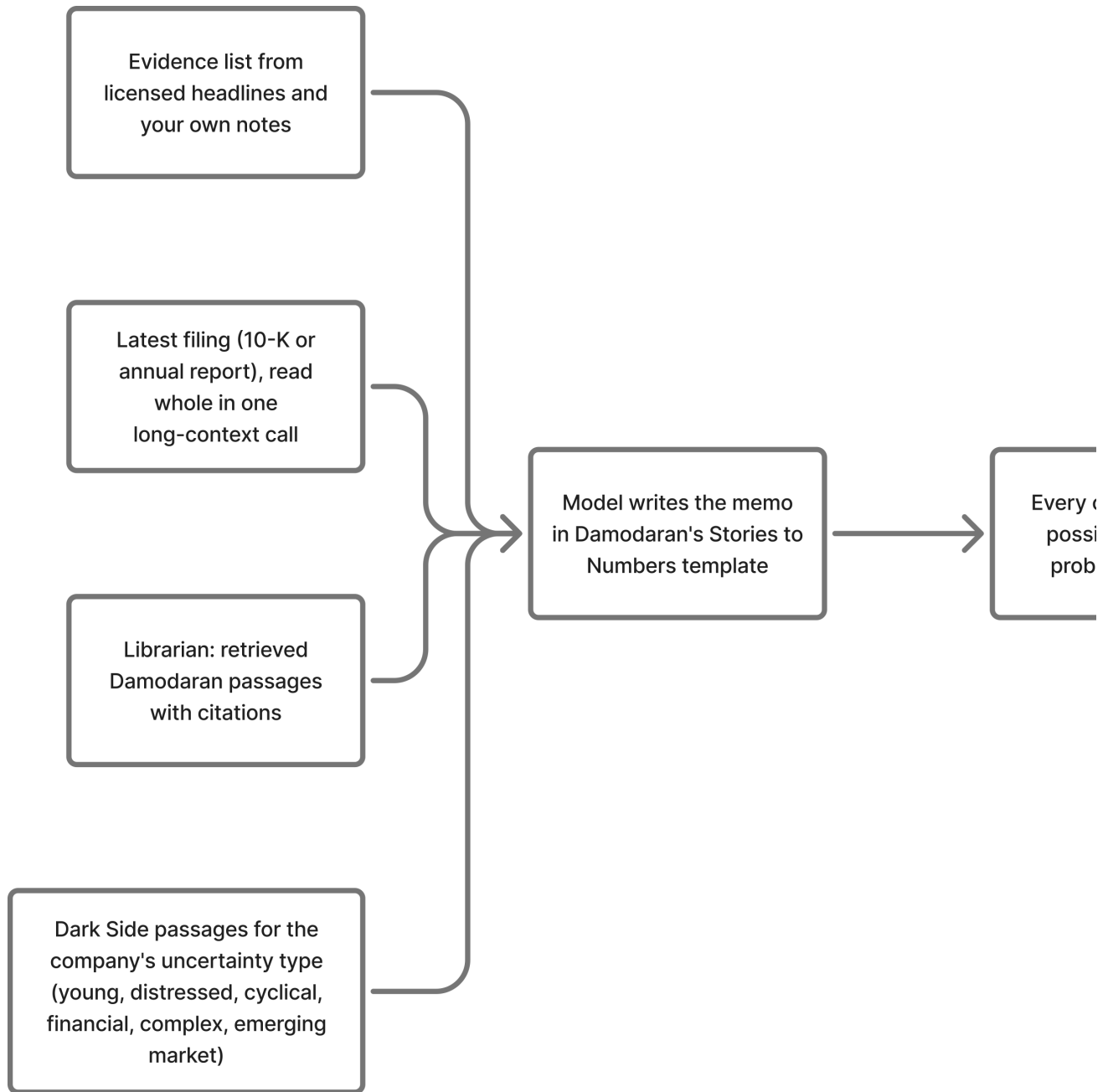


Figure 4: The memo pipeline

11 10. The accounts you will create, in order

Create them in this order, because each one's key is needed before the next part works. Use a different email alias for each vendor (you+neon@gmail.com, you+voyage@gmail.com) so a breach at one cannot be used to find the others. Turn on two-factor authentication everywhere.

#	Vendor	Plan	Cost per month	What you will copy, and where it goes	Notes
1	GitHub, plus a free organisation	Free	\$0	A fine-grained token scoped to one repo, 90-day expiry: GITHUB_TOKEN	Enable 2FA first and store recovery codes offline. The organisation is what keeps faster CI runners available later; personal accounts cannot use them.
2	Cursor (you have it)	Pro	\$20	Nothing	Settings: Privacy Mode on . Never paste an API key into Cursor.
3	OpenRouter	pay as you go	\$30–70	OPENROUTER_API_KEY	Buy \$20 of credit. Settings: monthly limit \$75 , prompt logging off , zero-data-retention routing on .
4	Neon	Free, later Launch	\$0, then \$10–20	DATABASE_URL (the “pooled” one), DATABASE_URL_UNPOOLED	Sign up with email and password, not only “continue with GitHub”.
5	Pinecone	Starter	\$0	PINECONE_API_KEY	On index, 1024 dimensions, cosine, region us-east-1.

#	Vendor	Plan	Cost per month	What you will copy, and where it goes	Notes
6	Voyage	Free	\$0	VOYAGE_API_KEY	Set a billing alert before the 200 million free tokens run out.
7	Cloudflare	Free (R2)	\$0	R2_ACCOUNT_ID, R2_ACCESS_KEY_ID, R2_SECRET_ACCESS_KEY, R2_BUCKET	Use a hardware key authenticator app; scope the token to one bucket.
8	Langfuse	Hobby	\$0	LANGFUSE_PUBLIC_KEY, LANGFUSE_SECRET_KEY, LANGFUSE_BASEURL	
9	Mistral	pay as you go	about \$2 once	MISTRAL_API_KEY	Used once for OCR through the batch endpoint.
10	Hex	Community	\$0, later \$36	The Neon connection is configured inside Hex (use the pooled host, SSL on)	Private workspace.
11	EODHD (milestone 4)	Fundamentals	\$60	EODHD_API_TOKEN	Test 20 NSE tickers on the free tier first; read the redistribution clause.
12	Optional later: INDmoney's INDstocks API, Kite Connect, Parallel search, Braintrust		\$0 / ₹500 / about \$18 / \$0		Broker tokens expire daily.

Where the keys go: a file called `.env` in the repo (never committed; GitHub's push protection refuses a commit that contains a key) and the "Secrets" page of the repo for the robots. A committed

`.env.example` lists the names with empty values.

12 11. Setting up Cursor for this repo

1. Settings, then General, then **Privacy Mode: on**. Your code is then not used for training, and Cursor's zero-data-retention agreements with the model providers apply.
2. Open the repo folder. The file `.cursor/mcp.json` (shipped with the repo) registers the system's tools. **MCP** stands for Model Context Protocol: a standard plug that lets an AI tool call other programs. You will see the tools listed in the agent panel: `value_company`, `write_memo`, `review_memo`, `get_industry_stats`, `search_corpus`, `report`.
3. The folder `.cursor/rules/` holds the hard rules (the six above, in one page). Cursor reads them on every request.
4. Press **Shift+Tab** to switch between Plan mode and Agent mode. In Plan mode the AI reads the code and writes a plan in English that you approve before anything changes. Use it for anything larger than a one-line edit.
5. Every change appears as a **diff**: the old lines in red, the new lines in green. Accept or reject each block. This is the single habit that keeps you the owner of the code.
6. Optional: install the **Cline** extension inside Cursor, give it your OpenRouter key, and set different models for its Plan and Act modes. That is your bench for feeling what an open-weight model does before its name goes into a robot.

13 12. Budget at up to \$200 a month

Line	Now	Notes
Cursor Pro	\$20 (already paying)	Coding stays on Cursor's included models
OpenRouter (the product's own model calls)	\$30–70	Spend limit \$75
EODHD (from milestone 4)	\$60	Personal-use licence
Hex	\$0, later \$36	When you want a published dashboard with daily refresh
GitHub private-repo minutes	\$0–25	2,000 free minutes, then \$0.006 a minute; spending cap \$25
Neon	\$0, later \$10–20	Free until 0.5 GB
Pinecone	\$0, later \$20	Free tier holds the whole corpus
Voyage, Langfuse, Cloudflare R2	\$0	Free at this size
Mistral OCR	about \$10 once	
Total	about \$120–200	Bloomberg Digital is your own subscription outside this

14 13. Bloomberg, INDmoney and news, honestly

- Your **bloomberg.com** subscription is a reading product. The “API library” page you found is the software kit for Terminal, Server API and Data License customers (the Terminal alone is

about \$32,000 a year). A digital subscriber gets no data access and no developer portal. What exists for free: headline-only RSS feeds and email newsletters. The feeds are a grey area under Bloomberg's terms, so the system stores only title, link and time, polls a few times a day, and never relies on them. Your own reading goes into short notes (link, date, your one sentence, the input it moves), never article text.

- **INDmoney** forbids scraping its app, but its broking arm publishes a free official API (INDstocks): your holdings, positions, quotes and 10 years of daily prices. Read-only endpoints work from a nightly job; order endpoints need a fixed IP address, which we never use.
- **Tavily and Firecrawl** are search and crawl services you pay per query; they are data sources, not schedulers. The scheduler is GitHub Actions. On the Artificial Analysis Search Index (18 August 2026), Parallel and Exa beat both on quality per dollar, so if we add news discovery it will be one of those.

15 14. What happens next

1. You read this document and “Damodaran, the important parts”.
2. You create the accounts in part 10 and set up Cursor as in part 11. Three evenings.
3. When you say go, I build the foundation in your private repo: calculator with the golden test, database, librarian, memo writer, tools, robots, first dashboard, and a walkthrough that explains every file.
4. You work in Cursor, one guide chapter and one small change at a time. I am on call.
5. India, the accuracy scoreboard, ranges and the market panel follow.

16 15. Glossary

- **Agent (AI)**. An AI model given tools (read files, run commands) and a loop, so it can carry out multi-step tasks rather than answer one question.
- **API**. A way for programs to talk to a service over the internet; “API key” is the password for it.
- **Branch (git)**. A parallel draft of the code. **Branch (Neon)**. A parallel copy of the database.
- **CI**. Continuous integration: every proposed change is tested automatically before it can be merged.
- **Chunk / passage**. A paragraph-sized slice of a document that the librarian indexes.
- **Cost per task**. What it costs, in dollars, for a model to complete one benchmark task; the number that maps to your bill, unlike price per token.
- **Cron / schedule**. A timer that runs a job at fixed times.
- **Diff**. The list of exact lines added and removed by a change.
- **Embedding / vector**. A list of numbers that places a passage in a space where similar meanings sit close together.
- **Golden test**. A test that pins a known-correct answer (Almarai) so any drift is caught.
- **Harness**. The scaffolding around a model: prompt, tools, loop, permissions, interface. Cursor and pi are harnesses.
- **Hash / fingerprint (sha256)**. A short code computed from a file's bytes; if the file changes, the code changes.
- **Licence class**. Our label for what you may do with a stored fact.
- **LTM**. Last twelve months of financials, built from the last annual report and interim reports.
- **MCP**. Model Context Protocol, the standard plug that lets Cursor call our tools.
- **Migration**. A versioned script that changes the database's structure.
- **Open-weight model**. A model whose weights are public; many hosts serve it.

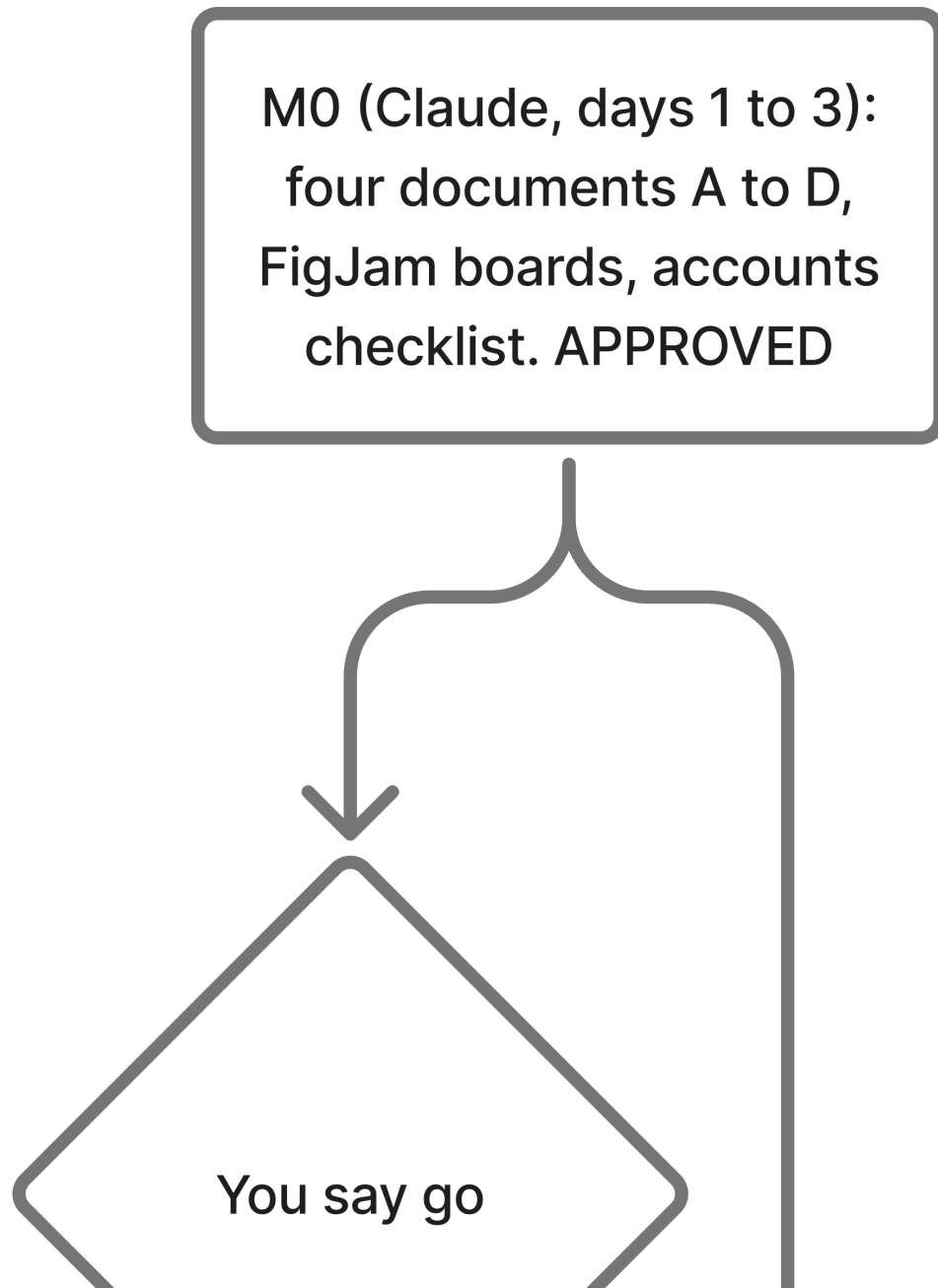


Figure 5: Milestone map

- **Reranker.** A second, slower model that re-scores the top passages for a question.
- **Repo.** A folder of code with its full history, hosted on GitHub.
- **Reverse DCF.** Solving the valuation backwards: what growth or margin does today's price require?
- **Sub-agent.** A helper AI session spawned by another to do a sub-task in its own context.
- **Vintage.** Which dated snapshot of Damodaran's tables a number came from.
- **Zero data retention.** A provider's promise not to store your prompts and outputs.